

# COIL\_Thanos

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## 1 Naive Bayes from Scratch

Curso: Minería de Datos en la Practica (UNAM)

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This notebook implements a Naive Bayes classifier from scratch to identify potential mobile home insurance purchasers using the COIL dataset. The analysis progresses from uniperspective driver identification through multiperspective pair analysis to customer scoring, achieving a 4.18x lift for the top 100 customers. The methodology translates raw insurance data into actionable marketing insights by isolating the most influential customer characteristics and their interactions.

### 1.1 Uniperspective Driver Analysis

Loads and preprocesses data, mapping data to enhance the descriptions, analyzes the features and calculated epilson values for each feature. Creates a dataframe with the most impactful drivers measured by epsilon.

```
[1]: import pandas as pd
import numpy as np
import math

# Load the dataset
df = pd.read_csv('tic_training.csv')

# Function to calculate Naive Bayes statistics for a feature-value pair
def calculate_stats(feature, value, description, subcategory,
    is_threshold=False):
    if is_threshold:
        condition = df[feature] >= value
        value_display = f" {value}"
        response = value_display
    else:
        condition = df[feature] == value
        value_display = value
        response = str(value)

# Count instances
N_X = condition.sum()
```

```

N_CX = ((df['CARAVAN'] == 1) & condition).sum()
N_not_CX = N_X - N_CX

# Calculate probabilities
P_C_given_X = N_CX / N_X if N_X > 0 else 0
prior_probability = df['CARAVAN'].mean()

# Calculate Epsilon
if N_X > 0 and prior_probability < 1 and prior_probability > 0:
    epsilon = (N_X * (P_C_given_X - prior_probability)) / (N_X *
prior_probability * (1 - prior_probability))**0.5
else:
    epsilon = 0

# Calculate Score
positive_instances = df['CARAVAN'].sum()
negative_instances = len(df) - positive_instances
P_X_given_C = N_CX / positive_instances if positive_instances > 0 else 0
P_X_given_not_C = N_not_CX / negative_instances if negative_instances > 0
else 0

if P_X_given_not_C > 0 and P_X_given_C > 0:
    score_X = math.log(P_X_given_C / P_X_given_not_C)
else:
    score_X = 0

s_r = math.log(prior_probability / (1 - prior_probability)) if
prior_probability < 1 and prior_probability > 0 else 0
score = score_X + s_r

return {
    'Subcategoría': subcategory,
    'Descripción': description,
    'Respuesta': response,
    'N(CX)': N_CX,
    'N(X)': N_X,
    'P(C|X)': round(P_C_given_X, 2),
    'P(C)': round(prior_probability, 2),
    'Epsilon': round(epsilon, 2),
    'Score': round(score, 2),
    'N(C)': positive_instances,
    'N': len(df),
    'N(-C)': negative_instances,
    'N(-CX)': N_not_CX
}

```

*# Complete mapping of field codes to their descriptions*

```

field_descriptions = {
    # Demographic data (1-43)
    'MOSTYPE': 'Customer Subtype',
    'MAANTHUI': 'Number of houses',
    'MGEMOMV': 'Average size household',
    'MGEMLEEF': 'Average age',
    'MOSHOOFD': 'Customer main type',
    'MGODRK': 'Roman catholic',
    'MGODPR': 'Protestant',
    'MGODOV': 'Other religion',
    'MGODGE': 'No religion',
    'MRELGE': 'Married',
    'MRELSA': 'Living together',
    'MRELOV': 'Other relation',
    'MFALLEEN': 'Singles',
    'MFGEKIND': 'Household without children',
    'MFWEKIND': 'Household with children',
    'MOPLHOOG': 'High level education',
    'MOPLMIDD': 'Medium level education',
    'MOPLLAAG': 'Lower level education',
    'MBERHOOG': 'High status',
    'MBERZELF': 'Entrepreneur',
    'MBERBOER': 'Farmer',
    'MBERMIDD': 'Middle management',
    'MBERARBG': 'Skilled laborers',
    'MBERARBO': 'Unskilled laborers',
    'MSKA': 'Social class A',
    'MSKB1': 'Social class B1',
    'MSKB2': 'Social class B2',
    'MSKC': 'Social class C',
    'MSKD': 'Social class D',
    'MHHUUR': 'Rented house',
    'MHKOOP': 'Home owners',
    'MAUT1': '1 car',
    'MAUT2': '2 cars',
    'MAUTO': 'No car',
    'MZFONDS': 'National Health Service',
    'MZPART': 'Private health insurance',
    'MINKM30': 'Income < 30.000',
    'MINK3045': 'Income 30-45.000',
    'MINK4575': 'Income 45-75.000',
    'MINK7512': 'Income 75-122.000',
    'MINK123M': 'Income >123.000',
    'MINKGEM': 'Average income',
    'MKOOPKLA': 'Purchasing power class',

    # Contribution (insurance premium payment) fields (44-64)

```

```

'PWAPART': 'Contribution private third party insurance',
'PWABEDR': 'Contribution third party insurance (firms)',
'PWALAND': 'Contribution third party insurance (agriculture)',
'PPERSAUT': 'Contribution car policies',
'PBESAUT': 'Contribution delivery van policies',
'PMOTSCO': 'Contribution motorcycle/scooter policies',
'PVRAAUT': 'Contribution lorry policies',
'PAANHANG': 'Contribution trailer policies',
'PTRACTOR': 'Contribution tractor policies',
'PWERKT': 'Contribution agricultural machines policies',
'PBROM': 'Contribution moped policies',
'PLEVEN': 'Contribution life insurances',
'PPERSONG': 'Contribution private accident insurance policies',
'PGEZONG': 'Contribution family accidents insurance policies',
'PWAOREG': 'Contribution disability insurance policies',
'PBRAND': 'Contribution fire policies',
'PZEILPL': 'Contribution surfboard policies',
'PPLEZIER': 'Contribution boat policies',
'PFIETS': 'Contribution bicycle policies',
'PINBOED': 'Contribution property insurance policies',
'PBYSTAND': 'Contribution social security insurance policies',

```

*# Policy count fields (65-85)*

```

'AWAPART': 'Number of private third party insurance',
'AWABEDR': 'Number of third party insurance (firms)',
'AWALAND': 'Number of third party insurance (agriculture)',
'APERSAUT': 'Number of car policies',
'ABESAUT': 'Number of delivery van policies',
'AMOTSCO': 'Number of motorcycle/scooter policies',
'AVRAAUT': 'Number of lorry policies',
'AAANHANG': 'Number of trailer policies',
'ATRACTOR': 'Number of tractor policies',
'AWERKT': 'Number of agricultural machines policies',
'ABROM': 'Number of moped policies',
'ALEVEN': 'Number of life insurances',
'APERSONG': 'Number of private accident insurance policies',
'AGEZONG': 'Number of family accidents insurance policies',
'AWAOREG': 'Number of disability insurance policies',
'ABRAND': 'Number of fire policies',
'AZEILPL': 'Number of surfboard policies',
'APLEZIER': 'Number of boat policies',
'AFIETS': 'Number of bicycle policies',
'AINBOED': 'Number of property insurance policies',
'ABYSTAND': 'Number of social security insurance policies',

```

*# Target variable*

```

'CARAVAN': 'Number of mobile home policies'

```

```

}

# Customer subtypes (MOSTYPE) from LO
customer_subtypes = {
    1: 'High Income, expensive child',
    2: 'Very Important Provincials',
    3: 'High status seniors',
    4: 'Affluent senior apartments',
    5: 'Mixed seniors',
    6: 'Career and childcare',
    7: "Dinki's (double income no kids)",
    8: 'Middle class families',
    9: 'Modern, complete families',
    10: 'Stable family',
    11: 'Family starters',
    12: 'Affluent young families',
    13: 'Young all american family',
    14: 'Junior cosmopolitan',
    15: 'Senior cosmopolitans',
    16: 'Students in apartments',
    17: 'Fresh masters in the city',
    18: 'Single youth',
    19: 'Suburban youth',
    20: 'Ethnically diverse',
    21: 'Young urban have-nots',
    22: 'Mixed apartment dwellers',
    23: 'Young and rising',
    24: 'Young, low educated',
    25: 'Young seniors in the city',
    26: 'Own home elderly',
    27: 'Seniors in apartments',
    28: 'Residential elderly',
    29: 'Porchless seniors: no front yard',
    30: 'Religious elderly singles',
    31: 'Low income catholics',
    32: 'Mixed seniors',
    33: 'Lower class large families',
    34: 'Large family, employed child',
    35: 'Village families',
    36: "Couples with teens 'Married with children'",
    37: 'Mixed small town dwellers',
    38: 'Traditional families',
    39: 'Large religious families',
    40: 'Large family farms',
    41: 'Mixed rurals'
}

```

```

# Age descriptions for MGEMLEEF (L1)
age_descriptions = {
    1: '20-30 years',
    2: '30-40 years',
    3: '40-50 years',
    4: '50-60 years',
    5: '60-70 years',
    6: '70-80 years'
}

# Customer main type descriptions for MOSHOOFD (L2)
customer_main_descriptions = {
    1: 'Successful hedonists',
    2: 'Driven Growers',
    3: 'Average Family',
    4: 'Career Loners',
    5: 'Living well',
    6: 'Cruising Seniors',
    7: 'Retired and Religious',
    8: 'Family with grown ups',
    9: 'Conservative families',
    10: 'Farmers'
}

# Percentage descriptions (L3)
percentage_descriptions = {
    0: '0%',
    1: '1-10%',
    2: '11-23%',
    3: '24-36%',
    4: '37-49%',
    5: '50-62%',
    6: '63-75%',
    7: '76-88%',
    8: '89-99%',
    9: '100%'
}

# Contribution amounts (L4)
contribution_descriptions = {
    0: '0',
    1: '1-49',
    2: '50-99',
    3: '100-199',
    4: '200-499',
    5: '500-999',
    6: '1000-4999',

```

```

7: '5000-9999',
8: '10,000-19,999',
9: '20,000+'
}

# Function to enhance driver descriptions with meaningful labels
def enhance_description(description):
    parts = description.split(' = ')
    if len(parts) == 2:
        field, value = parts
        try:
            value = int(value)

            if field in field_descriptions:
                field_desc = field_descriptions[field]

                # Apply specific value descriptions based on field type
                if field == 'MOSTYPE' and value in customer_subtypes:
                    value_desc = customer_subtypes[value]
                elif field == 'MGEMLEEF' and value in age_descriptions:
                    value_desc = age_descriptions[value]
                elif field == 'MOSHOOFD' and value in_
↳customer_main_descriptions:
                    value_desc = customer_main_descriptions[value]
                # Percentage fields (many sociodemographic variables)
                elif field.startswith('M') and value in percentage_descriptions:
                    value_desc = percentage_descriptions[value]
                # Contribution fields
                elif field.startswith('P') and value in_
↳contribution_descriptions:
                    value_desc = contribution_descriptions[value]
                # Count fields (policy counts)
                elif field.startswith('A'):
                    value_desc = str(value) + " policies"
                else:
                    value_desc = str(value)

            return f"{field_desc}: {value_desc}"
        except ValueError:
            pass

# For thresholds (e.g., MINKGEM >= 4)
if '>=' in description:
    parts = description.split(' >= ')
    if len(parts) == 2:
        field, threshold = parts
        try:

```

```

        threshold = int(threshold)

        if field in field_descriptions:
            field_desc = field_descriptions[field]

            # Special case for income thresholds
            if field == 'MINKGEM':
                return f"{field_desc} >= level {threshold}"
            elif field == 'MKOOPKLA':
                return f"{field_desc} >= class {threshold}"
            else:
                return f"{field_desc} >= {threshold}"
        except ValueError:
            pass

    # Default: return the original description
    return description

# List of features to analyze
features_to_analyze = [
    # Demographics
    'MOSTYPE', 'MOSHOOFD', 'MGEMLEEF', 'MAANTHUI', 'MGEMOMV',
    # Religion/relationships
    'MGODRK', 'MGODPR', 'MGODOV', 'MGODGE', 'MRELGE', 'MRELSA', 'MRELOV',
    # Family status
    'MFALLEEN', 'MFGEKIND', 'MFWEKIND',
    # Education
    'MOPLHOOG', 'MOPLMIDD', 'MOPLLAAG',
    # Professional status
    'MBERHOOG', 'MBERZELF', 'MBERBOER', 'MBERMIDD', 'MBERARBG', 'MBERARBO',
    # Social class
    'MSKA', 'MSKB1', 'MSKB2', 'MSKC', 'MSKD',
    # Housing
    'MHHUUR', 'MHKOOP',
    # Transportation
    'MAUT1', 'MAUT2', 'MAUTO',
    # Insurance
    'MZFONDS', 'MZPART',
    # Income
    'MINKM30', 'MINK3045', 'MINK4575', 'MINK7512', 'MINK123M', 'MINKGEM',
    ↪ 'MKOOPKLA',
    # Insurance contributions
    'PWAPART', 'PWABEDR', 'PWALAND', 'PPERSAUT', 'PBESAUT', 'PMOTSCO',
    'PVRAAUT', 'PAANHANG', 'PTRACTOR', 'PWERKT', 'PBROM', 'PLEVEN',
    'PPERSONG', 'PGEZONG', 'PWAOREG', 'PBRAND', 'PZEILPL', 'PPLEZIER',
    'PFIETS', 'PINBOED', 'PBYSTAND',
    # Number of policies

```

```

    'AWAPART', 'AWABEDR', 'AWALAND', 'APERSAUT', 'ABESAUT', 'AMOTSCO',
    'AVRAAUT', 'AAANHANG', 'ATTRACTOR', 'AWERKT', 'ABROM', 'ALEVEN',
    'APERSONG', 'AGEZONG', 'AWAOREG', 'ABRAND', 'AZEILPL', 'APLEZIER',
    'AFIETS', 'AINBOED', 'ABYSTAND'
]

# Initialize results list
results = []

# Process each feature
for feature in features_to_analyze:
    # Determine category based on feature type
    if feature.startswith('M'):
        if feature in ['MOSTYPE', 'MOSHOOFD', 'MGEMLEEF', 'MAANTHUI',
↪ 'MGEMOMV']:
            category = 'Demographics'
        elif feature in ['MGODRK', 'MGODPR', 'MGODOV', 'MGODGE']:
            category = 'Religion'
        elif feature in ['MRELGE', 'MRELSA', 'MRELOV', 'MFALLEEN', 'MFGEKIND',
↪ 'MFWEKIND']:
            category = 'Family Status'
        elif feature in ['MOPLHOOG', 'MOPLMIDD', 'MOPLLAAG']:
            category = 'Education'
        elif feature in ['MBERHOOG', 'MBERZELF', 'MBERBOER', 'MBERMIDD',
↪ 'MBERARBG', 'MBERARBO']:
            category = 'Professional Status'
        elif feature in ['MSKA', 'MSKB1', 'MSKB2', 'MSKC', 'MSKD']:
            category = 'Social Class'
        elif feature in ['MHUUR', 'MHKOOP']:
            category = 'Housing'
        elif feature in ['MAUT1', 'MAUT2', 'MAUTO']:
            category = 'Transportation'
        elif feature in ['MZFONDS', 'MZPART']:
            category = 'Health Insurance'
        elif feature in ['MINKM30', 'MINK3045', 'MINK4575', 'MINK7512',
↪ 'MINK123M', 'MINKGEM', 'MKOOPKLA']:
            category = 'Income'
        else:
            category = 'Sociodemographic'
    elif feature.startswith('P'):
        category = 'Insurance Contributions'
    elif feature.startswith('A'):
        category = 'Policy Counts'
    else:
        category = 'Other'

# Analyze each unique value for the feature

```

```

unique_values = sorted(df[feature].unique())
for value in unique_values:
    description = f"{feature} = {value}"
    results.append(calculate_stats(feature, value, description, category))

# For income and purchasing power features, also analyze threshold values
if feature in ['MINKGEM', 'MKOOPKLA']:
    for threshold in range(1, 10):
        description = f"{feature} >= {threshold}"
        results.append(calculate_stats(feature, threshold, description,
↪ 'Income', is_threshold=True))

# Create DataFrame with results
results_df = pd.DataFrame(results)

# Sort by absolute epsilon value to find most significant factors
results_df = results_df.sort_values(by='Epsilon', ascending=False)

# Create the final output with enhanced descriptions and corrected percentage
↪ calculation
top_drivers = results_df.head(20)
output_df = pd.DataFrame({
    'Driver': top_drivers['Descripción'].apply(enhance_description),
    'Category': top_drivers['Subcategoría'],
    'Epsilon': top_drivers['Epsilon'],
    'Customers with Driver': top_drivers['N(X)'],
    'Purchasers with Driver': top_drivers['N(CX)'],
    # Calculate percentage directly from raw counts to avoid double rounding
    'Percentage (%)': ((top_drivers['N(CX)'] / top_drivers['N(X)']) * 100).
↪ round(2)
})

# Display the results
print("\nTop Uniperspective Drivers for Mobile Home Insurance:")
print(output_df)

```

Top Uniperspective Drivers for Mobile Home Insurance:

|     | Driver \                                |
|-----|-----------------------------------------|
| 469 | Contribution car policies: 1000-4999    |
| 537 | Contribution fire policies: 200-499     |
| 634 | Number of boat policies: 1 policies     |
| 440 | Purchasing power class: 76-88%          |
| 7   | Customer Subtype: Middle class families |
| 448 | Purchasing power class >= class 7       |
| 41  | Customer main type: Driven Growers      |
| 576 | Number of car policies: 2 policies      |

575                   Number of car policies: 1 policies  
 447                   Purchasing power class >= class 6  
 453   Contribution private third party insurance: 50-99  
 567   Number of private third party insurance: 1 pol...  
 644   Number of social security insurance policies: ...  
 428                   Average income >= level 4  
 546                   Contribution boat policies: 1-49  
 318                   Home owners: 100%  
 299                   Rented house: 0%  
 446                   Purchasing power class >= class 5  
 429                   Average income >= level 5  
 477   Contribution motorcycle/scooter policies: 100-199

|     | Category                | Epsilon | Customers with Driver \ |
|-----|-------------------------|---------|-------------------------|
| 469 | Insurance Contributions | 10.81   | 2319                    |
| 537 | Insurance Contributions | 9.36    | 1226                    |
| 634 | Policy Counts           | 7.69    | 31                      |
| 440 | Income                  | 7.49    | 474                     |
| 7   | Demographics            | 7.04    | 339                     |
| 448 | Income                  | 6.78    | 900                     |
| 41  | Demographics            | 6.78    | 502                     |
| 576 | Policy Counts           | 6.27    | 246                     |
| 575 | Policy Counts           | 6.07    | 2712                    |
| 447 | Income                  | 6.00    | 1801                    |
| 453 | Insurance Contributions | 5.83    | 2128                    |
| 567 | Policy Counts           | 5.37    | 2334                    |
| 644 | Policy Counts           | 5.23    | 81                      |
| 428 | Income                  | 5.16    | 3165                    |
| 546 | Insurance Contributions | 5.10    | 5                       |
| 318 | Housing                 | 5.10    | 949                     |
| 299 | Housing                 | 5.10    | 949                     |
| 446 | Income                  | 4.79    | 2384                    |
| 429 | Income                  | 4.73    | 1311                    |
| 477 | Insurance Contributions | 4.43    | 3                       |

|     | Purchasers with Driver | Percentage (%) |
|-----|------------------------|----------------|
| 469 | 262                    | 11.30          |
| 537 | 151                    | 12.32          |
| 634 | 12                     | 38.71          |
| 440 | 67                     | 14.14          |
| 7   | 51                     | 15.04          |
| 448 | 102                    | 11.33          |
| 41  | 66                     | 13.15          |
| 576 | 38                     | 15.45          |
| 575 | 237                    | 8.74           |
| 447 | 168                    | 9.33           |
| 453 | 191                    | 8.98           |
| 567 | 201                    | 8.61           |

|     |     |       |
|-----|-----|-------|
| 644 | 16  | 19.75 |
| 428 | 258 | 8.15  |
| 546 | 3   | 60.00 |
| 318 | 94  | 9.91  |
| 299 | 94  | 9.91  |
| 446 | 198 | 8.31  |
| 429 | 119 | 9.08  |
| 477 | 2   | 66.67 |

## 1.2 Multiperspective approach with 2 drivers

Uses a beam search approach to find the top pairs of individual drives. Iterates through all pairs of the top 40 predictors to find the most promising ones.

```
[ ]: # Function to calculate statistics for a pair of drivers
def calculate_pair_stats(feature1, value1, feature2, value2,
    is_threshold1=False, is_threshold2=False):
    # Create conditions for both features
    if is_threshold1:
        condition1 = df[feature1] >= value1
    else:
        condition1 = df[feature1] == value1

    if is_threshold2:
        condition2 = df[feature2] >= value2
    else:
        condition2 = df[feature2] == value2

    # Combined condition (logical AND)
    combined_condition = condition1 & condition2

    # Count instances
    N_X = combined_condition.sum()
    N_CX = ((df['CARAVAN'] == 1) & combined_condition).sum()
    N_not_CX = N_X - N_CX

    # Calculate probabilities
    P_C_given_X = N_CX / N_X if N_X > 0 else 0
    prior_probability = df['CARAVAN'].mean()

    # Calculate Epsilon
    if N_X > 0 and prior_probability < 1 and prior_probability > 0:
        epsilon = (N_X * (P_C_given_X - prior_probability)) / (N_X *
            prior_probability * (1 - prior_probability))*0.5
    else:
        epsilon = 0

    # Get descriptions
```

```

description1 = f"{feature1} = {value1}" if not is_threshold1 else
↪ f"{feature1} >= {value1}"
description2 = f"{feature2} = {value2}" if not is_threshold2 else
↪ f"{feature2} >= {value2}"
combined_description = f"{description1} AND {description2}"

# Determine category
categories = []
for feature in [feature1, feature2]:
    if feature.startswith('M'):
        if feature in ['MOSTYPE', 'MOSHOOFD', 'MGEMLEEF', 'MAANTHUI',
↪ 'MGEMOMV']:
            categories.append('Demographics')
        elif feature in ['MGODRK', 'MGODPR', 'MGODOV', 'MGODGE']:
            categories.append('Religion')
        elif feature in ['MRELGE', 'MRELSA', 'MRELOV', 'MFALLEEN',
↪ 'MFGEKIND', 'MFWEKIND']:
            categories.append('Family Status')
        elif feature in ['MOPLHOOG', 'MOPLMIDD', 'MOPLLAAG']:
            categories.append('Education')
        elif feature in ['MBERHOOG', 'MBERZELF', 'MBERBOER', 'MBERMIDD',
↪ 'MBERARBG', 'MBERARBO']:
            categories.append('Professional Status')
        else:
            categories.append('Sociodemographic')
    elif feature.startswith('P'):
        categories.append('Insurance Contributions')
    elif feature.startswith('A'):
        categories.append('Policy Counts')
    else:
        categories.append('Other')

category = ' + '.join(sorted(set(categories)))

return {
    'Description': combined_description,
    'Category': category,
    'N(X)': N_X,
    'N(CX)': N_CX,
    'Epsilon': round(epsilon, 2),
    'Driver1': description1,
    'Driver2': description2,
    'Is_Threshold1': is_threshold1,
    'Is_Threshold2': is_threshold2,
    'Feature1': feature1,
    'Value1': value1,
    'Feature2': feature2,

```

```

        'Value2': value2
    }

# Get top uniperspective drivers for beam search
top_n = 40 # Adjust based on computational capacity, focus only on top 40
↳ drivers
top_uniperspective = results_df.sort_values(by='Epsilon', ascending=False).
↳ head(top_n)

# Initialize list for multiperspective results
multiperspective_results = []

# Generate and evaluate pairs (using beam search approach)
print("Searching for multiperspective drivers...")

# Extract feature-value pairs from top uniperspective drivers
driver_pairs = []
for i, row1 in top_uniperspective.iterrows():
    desc1 = row1['Descripción']

    # Check if it's a threshold condition
    is_threshold1 = '>=' in desc1
    if is_threshold1:
        parts = desc1.split(' >= ')
        feature1 = parts[0]
        value1 = int(parts[1])
    else:
        parts = desc1.split(' = ')
        feature1 = parts[0]
        value1 = int(parts[1])

    for j, row2 in top_uniperspective.iterrows():
        if i >= j: # Avoid duplicates and same-feature pairs
            continue

        desc2 = row2['Descripción']

        # Check if it's a threshold condition
        is_threshold2 = '>=' in desc2
        if is_threshold2:
            parts = desc2.split(' >= ')
            feature2 = parts[0]
            value2 = int(parts[1])
        else:
            parts = desc2.split(' = ')
            feature2 = parts[0]
            value2 = int(parts[1])

```

```

    # Avoid pairs of the same feature
    if feature1 == feature2:
        continue

    # Calculate stats for this pair
    pair_stats = calculate_pair_stats(
        feature1, value1, feature2, value2,
        is_threshold1=is_threshold1,
        is_threshold2=is_threshold2
    )

    # Only consider pairs with sufficient support (at least 30 customers)
    if pair_stats['N(X)'] >= 30:
        multiperspective_results.append(pair_stats)

# Convert to DataFrame and sort by epsilon
multiperspective_df = pd.DataFrame(multiperspective_results)
multiperspective_df = multiperspective_df.sort_values(by='Epsilon',
    ↪ascending=False)

# Enhance descriptions for the top pairs
def enhance_pair_description(row):
    # Enhance description for first driver
    if row['Is_Threshold1']:
        parts = row['Driver1'].split(' >= ')
        feature = parts[0]
        value = int(parts[1])
        enhanced1 = enhance_description(f"{feature} >= {value}")
    else:
        parts = row['Driver1'].split(' = ')
        feature = parts[0]
        value = int(parts[1])
        enhanced1 = enhance_description(f"{feature} = {value}")

    # Enhance description for second driver
    if row['Is_Threshold2']:
        parts = row['Driver2'].split(' >= ')
        feature = parts[0]
        value = int(parts[1])
        enhanced2 = enhance_description(f"{feature} >= {value}")
    else:
        parts = row['Driver2'].split(' = ')
        feature = parts[0]
        value = int(parts[1])
        enhanced2 = enhance_description(f"{feature} = {value}")

```

```

return f"{enhanced1} AND {enhanced2}"

# Create final output DataFrame with enhanced descriptions
top_pairs = multiperspective_df.head(20)
output_pairs_df = pd.DataFrame({
    'Driver Combination': top_pairs.apply(enhance_pair_description, axis=1),
    'Category': top_pairs['Category'],
    'Epsilon': top_pairs['Epsilon'],
    'Customers with Both Drivers': top_pairs['N(X)'],
    'Purchasers with Both Drivers': top_pairs['N(CX)'],
    'Percentage (%)': ((top_pairs['N(CX)'] / top_pairs['N(X)']) * 100).round(2)
})

# Display the results
print("\nTop Multiperspective Drivers for Mobile Home Insurance:")
print(output_pairs_df)

```

Searching for multiperspective drivers...

Top Multiperspective Drivers for Mobile Home Insurance:

|     | Driver Combination \                              | Category                                   | Epsilon \ |
|-----|---------------------------------------------------|--------------------------------------------|-----------|
| 77  | Purchasing power class >= class 6 AND Contribu... | Insurance Contributions + Sociodemographic | 13.53     |
| 0   | Contribution car policies: 1000-4999 AND Contr... | Insurance Contributions                    | 13.48     |
| 96  | Average income >= level 4 AND Contribution car... | Insurance Contributions + Sociodemographic | 13.30     |
| 144 | Purchasing power class >= class 5 AND Contribu... | Insurance Contributions + Sociodemographic | 12.78     |
| 5   | Contribution car policies: 1000-4999 AND Numbe... | Insurance Contributions + Policy Counts    | 12.61     |
| 41  | Purchasing power class >= class 7 AND Contribu... |                                            |           |
| 85  | Contribution private third party insurance: 50... |                                            |           |
| 3   | Contribution car policies: 1000-4999 AND Numbe... |                                            |           |
| 384 | Purchasing power class >= class 4 AND Contribu... |                                            |           |
| 97  | Average income >= level 4 AND Contribution fir... |                                            |           |
| 48  | Customer main type: Driven Growers AND Contrib... |                                            |           |
| 78  | Purchasing power class >= class 6 AND Contribu... |                                            |           |
| 385 | Purchasing power class >= class 4 AND Contribu... |                                            |           |
| 145 | Purchasing power class >= class 5 AND Contribu... |                                            |           |
| 126 | Rented house: 0% AND Contribution car policies... |                                            |           |
| 109 | Home owners: 100% AND Contribution car policie... |                                            |           |
| 7   | Contribution fire policies: 200-499 AND Number... |                                            |           |
| 42  | Purchasing power class >= class 7 AND Contribu... |                                            |           |
| 152 | Average income >= level 5 AND Contribution car... |                                            |           |
| 11  | Purchasing power class: 76-88% AND Contributio... |                                            |           |

|     |                                            |       |
|-----|--------------------------------------------|-------|
| 41  | Insurance Contributions + Sociodemographic | 12.50 |
| 85  | Insurance Contributions                    | 12.26 |
| 3   | Insurance Contributions + Policy Counts    | 12.19 |
| 384 | Insurance Contributions + Sociodemographic | 12.19 |
| 97  | Insurance Contributions + Sociodemographic | 11.89 |
| 48  | Demographics + Insurance Contributions     | 11.26 |
| 78  | Insurance Contributions + Sociodemographic | 11.25 |
| 385 | Insurance Contributions + Sociodemographic | 11.13 |
| 145 | Insurance Contributions + Sociodemographic | 11.13 |
| 126 | Insurance Contributions + Sociodemographic | 11.01 |
| 109 | Insurance Contributions + Sociodemographic | 11.01 |
| 7   | Insurance Contributions + Policy Counts    | 10.99 |
| 42  | Insurance Contributions + Sociodemographic | 10.88 |
| 152 | Insurance Contributions + Sociodemographic | 10.71 |
| 11  | Insurance Contributions + Sociodemographic | 10.65 |

|     | Customers with Both Drivers | Purchasers with Both Drivers | Percentage (%) |
|-----|-----------------------------|------------------------------|----------------|
| 77  | 761                         | 134                          | 17.61          |
| 0   | 680                         | 124                          | 18.24          |
| 96  | 1335                        | 195                          | 14.61          |
| 144 | 985                         | 154                          | 15.63          |
| 5   | 1315                        | 187                          | 14.22          |
| 41  | 384                         | 81                           | 21.09          |
| 85  | 1070                        | 159                          | 14.86          |
| 3   | 1143                        | 166                          | 14.52          |
| 384 | 1311                        | 183                          | 13.96          |
| 97  | 804                         | 128                          | 15.92          |
| 48  | 215                         | 52                           | 24.19          |
| 78  | 555                         | 96                           | 17.30          |
| 385 | 824                         | 125                          | 15.17          |
| 145 | 694                         | 111                          | 15.99          |
| 126 | 407                         | 77                           | 18.92          |
| 109 | 407                         | 77                           | 18.92          |
| 7   | 684                         | 109                          | 15.94          |
| 42  | 288                         | 61                           | 21.18          |
| 152 | 564                         | 94                           | 16.67          |
| 11  | 216                         | 50                           | 23.15          |

### 1.3 Using the score to identify predictors

Implements a predictive model using naive bayes to score each customer on their likelihood to purchase mobile home insurance.

Feature selection: search top 10 multiperspective pairs (based on epsilon) and include non-redundant feature pairs in scoring model. Then, take top 20 uniperspective drivers and include features that were not already used in pairs.

Shows that in this case, only 14 features were selected for the model.

```

[3]: import numpy as np
import pandas as pd
import math
from tqdm import tqdm

def calculate_customer_scores(df, top_drivers, top_pairs):
    print("Calculating scores for all customers...")

    # Get prior probability of having a caravan policy
    prior_probability = df['CARAVAN'].mean()
    prior_odds = prior_probability / (1 - prior_probability)
    log_prior = math.log(prior_odds) if prior_odds > 0 else 0

    # Initialize array for scores
    customer_scores = np.ones(len(df)) * log_prior

    # Dictionary to store likelihood ratios for feature-value pairs
    likelihood_ratios = {}

    # Track features used to avoid duplicate counting
    used_features = set()
    selected_pairs = []
    selected_single_features = []

    # Track features used to avoid duplicate counting
    used_features = set()

    # First use the top multiperspective pairs (stronger signal)
    for _, pair in top_pairs.iterrows():
        feature1 = pair['Feature1']
        value1 = pair['Value1']
        is_threshold1 = pair['Is_Threshold1']

        feature2 = pair['Feature2']
        value2 = pair['Value2']
        is_threshold2 = pair['Is_Threshold2']

        # Skip if either feature already used
        if feature1 in used_features or feature2 in used_features:
            continue

        # Add both features to the used set
        used_features.add(feature1)
        used_features.add(feature2)

        # Record the selected pair
        pair_desc = {

```

```

        'feature1': feature1,
        'value1': value1,
        'is_threshold1': is_threshold1,
        'feature2': feature2,
        'value2': value2,
        'is_threshold2': is_threshold2
    }
    selected_pairs.append(pair_desc)

    # Calculate likelihood ratios for this pair
    if is_threshold1:
        condition1 = df[feature1] >= value1
    else:
        condition1 = df[feature1] == value1

    if is_threshold2:
        condition2 = df[feature2] >= value2
    else:
        condition2 = df[feature2] == value2

    condition = condition1 & condition2

    # Count customers with and without caravan policies
    positive_instances = df['CARAVAN'].sum()
    negative_instances = len(df) - positive_instances

    # Calculate conditional probabilities with Laplace smoothing
    count_with_C = ((df['CARAVAN'] == 1) & condition).sum() + 1
    count_with_not_C = (condition & (df['CARAVAN'] == 0)).sum() + 1

    P_X_given_C = count_with_C / (positive_instances + 2)
    P_X_given_not_C = count_with_not_C / (negative_instances + 2)

    # Calculate log likelihood ratio for this pair
    log_ratio = math.log(P_X_given_C / P_X_given_not_C)

    # Apply to matching customers
    customer_scores[condition] += log_ratio

    # For customers that don't match this pair, apply the complement
    ↪likelihood
    comp_count_with_C = positive_instances - ((df['CARAVAN'] == 1) &
    ↪condition).sum() + 1
    comp_count_with_not_C = negative_instances - (condition &
    ↪(df['CARAVAN'] == 0)).sum() + 1
    comp_P_X_given_C = comp_count_with_C / (positive_instances + 2)
    comp_P_X_given_not_C = comp_count_with_not_C / (negative_instances + 2)

```

```

comp_log_ratio = math.log(comp_P_X_given_C / comp_P_X_given_not_C)

customer_scores[~condition] += comp_log_ratio

# Then add top uniperspective drivers for features not yet used
for _, row in top_drivers.iterrows():
    desc = row['Descripción']
    is_threshold = '>=' in desc

    if is_threshold:
        parts = desc.split(' >= ')
        feature = parts[0]
        value = int(parts[1])
    else:
        parts = desc.split(' = ')
        feature = parts[0]
        value = int(parts[1])

    # Skip if feature already used in pairs
    if feature in used_features:
        continue

    # Add feature to used set
    used_features.add(feature)

    # Record the selected single feature
    feature_desc = {
        'feature': feature,
        'value': value,
        'is_threshold': is_threshold
    }
    selected_single_features.append(feature_desc)

    # Create condition
    if is_threshold:
        condition = df[feature] >= value
    else:
        condition = df[feature] == value

    # Count customers with and without caravan policies
    positive_instances = df['CARAVAN'].sum()
    negative_instances = len(df) - positive_instances

    # Calculate conditional probabilities with Laplace smoothing
    count_with_C = ((df['CARAVAN'] == 1) & condition).sum() + 1
    count_with_not_C = (condition & (df['CARAVAN'] == 0)).sum() + 1

```

```

P_X_given_C = count_with_C / (positive_instances + 2)
P_X_given_not_C = count_with_not_C / (negative_instances + 2)

# Calculate log likelihood ratio
log_ratio = math.log(P_X_given_C / P_X_given_not_C)

# Apply to matching customers
customer_scores[condition] += log_ratio

# For customers that don't match, apply the complement likelihood
comp_count_with_C = positive_instances - ((df['CARAVAN'] == 1) &
↳condition).sum() + 1
comp_count_with_not_C = negative_instances - (condition &
↳(df['CARAVAN'] == 0)).sum() + 1
comp_P_X_given_C = comp_count_with_C / (positive_instances + 2)
comp_P_X_given_not_C = comp_count_with_not_C / (negative_instances + 2)
comp_log_ratio = math.log(comp_P_X_given_C / comp_P_X_given_not_C)

customer_scores[~condition] += comp_log_ratio

# Create DataFrame with customer IDs and scores
scores_df = pd.DataFrame({
    'CustomerID': df['NID'],
    'Score': customer_scores,
    'Probability': 1 / (1 + np.exp(-customer_scores)), # Convert log-odds
↳to probability
    'Actual': df['CARAVAN']
})

# Sort by score in descending order
scores_df = scores_df.sort_values(by='Score', ascending=False)

# Calculate lift for the top N groups
for n in [100, 200, 300, 500, 1000]:
    if len(scores_df) >= n:
        top_n = scores_df.head(n)
        lift = (top_n['Actual'].mean() / prior_probability)
        print(f"Lift for top {n} customers: {lift:.2f}x")

return scores_df, used_features, selected_pairs, selected_single_features

# Calculate scores using both uniperspective and multiperspective drivers
top_uniperspective_for_scoring = results_df.sort_values(by='Epsilon',
↳ascending=False).head(20)
top_multiperspective_for_scoring = multiperspective_df.head(10)

```

```

# Calculate scores and get selected features
customer_scores, used_features, selected_pairs, selected_single_features = calculate_customer_scores(
    df, top_uniperspective_for_scoring, top_multiperspective_for_scoring
)

# Display selected features
print("\n=== FEATURES SELECTED FOR THE FINAL MODEL ===")

print("\nSelected Feature Pairs:")
for i, pair in enumerate(selected_pairs, 1):
    # Format the first feature
    if pair['is_threshold1']:
        feature1_desc = f"{pair['feature1']} >= {pair['value1']}"
    else:
        feature1_desc = f"{pair['feature1']} = {pair['value1']}"

    # Format the second feature
    if pair['is_threshold2']:
        feature2_desc = f"{pair['feature2']} >= {pair['value2']}"
    else:
        feature2_desc = f"{pair['feature2']} = {pair['value2']}"

    # Get readable descriptions if possible
    try:
        feature1_readable = enhance_description(feature1_desc)
        feature2_readable = enhance_description(feature2_desc)
        print(f"{i}. {feature1_readable} AND {feature2_readable}")
    except:
        print(f"{i}. {feature1_desc} AND {feature2_desc}")

print("\nSelected Individual Features:")
for i, feature in enumerate(selected_single_features, 1):
    if feature['is_threshold']:
        feature_desc = f"{feature['feature']} >= {feature['value']}"
    else:
        feature_desc = f"{feature['feature']} = {feature['value']}"

    # Get readable description if possible
    try:
        feature_readable = enhance_description(feature_desc)
        print(f"{i}. {feature_readable}")
    except:
        print(f"{i}. {feature_desc}")

print(f"\nTotal number of unique features used: {len(used_features)}")
print("Feature list:", ", ".join(sorted(used_features)))

```

```

# Display the top 20 customers most likely to purchase
print("\nTop 20 Customers Most Likely to Purchase Mobile Home Insurance:")
top_customers = customer_scores.head(20)
print(top_customers[['CustomerID', 'Score', 'Probability', 'Actual']])

# Calculate performance metrics
total_positive = df['CARAVAN'].sum()
print(f"\nTotal customers with caravan policies: {total_positive}")
print(f"Baseline probability: {(total_positive / len(df)) * 100:.2f}%")

# Calculate precision at different thresholds
for k in [100, 200, 500, 1000]:
    if len(customer_scores) >= k:
        precision = customer_scores.head(k)['Actual'].sum() / k
        recall = customer_scores.head(k)['Actual'].sum() / total_positive
        print(f"Precision@{k}: {precision*100:.2f}%, Recall@{k}: {recall*100:.2f}%")

```

Calculating scores for all customers...

Lift for top 100 customers: 4.18x  
 Lift for top 200 customers: 4.77x  
 Lift for top 300 customers: 4.24x  
 Lift for top 500 customers: 3.81x  
 Lift for top 1000 customers: 2.74x

=== FEATURES SELECTED FOR THE FINAL MODEL ===

Selected Feature Pairs:

1. Purchasing power class >= class 6 AND Contribution car policies: 1000-4999
2. Average income >= level 4 AND Contribution fire policies: 200-499

Selected Individual Features:

1. Number of boat policies: 1 policies
2. Customer Subtype: Middle class families
3. Customer main type: Driven Growers
4. Number of car policies: 2 policies
5. Contribution private third party insurance: 50-99
6. Number of private third party insurance: 1 policies
7. Number of social security insurance policies: 1 policies
8. Contribution boat policies: 1-49
9. Home owners: 100%
10. Rented house: 0%

Total number of unique features used: 14

Feature list: ABYSTAND, APERSAUT, APLEZIER, AWAPART, MHHUUR, MHKOOP, MINKGEM, MKOOPKLA, MOSHOOFD, MOSTYPE, PBRAND, PPERSAUT, PPLEZIER, PWAPART

Top 20 Customers Most Likely to Purchase Mobile Home Insurance:

|      | CustomerID | Score    | Probability | Actual |
|------|------------|----------|-------------|--------|
| 2047 | 2048       | 4.672685 | 0.990739    | 0      |
| 4712 | 4713       | 4.663669 | 0.990656    | 0      |
| 2188 | 2189       | 4.444219 | 0.988390    | 0      |
| 1671 | 1672       | 4.417333 | 0.988077    | 1      |
| 4450 | 4451       | 4.394178 | 0.987802    | 0      |
| 2095 | 2096       | 4.382899 | 0.987665    | 1      |
| 5471 | 5472       | 4.382899 | 0.987665    | 0      |
| 5136 | 5137       | 4.382899 | 0.987665    | 1      |
| 3902 | 3903       | 4.235901 | 0.985740    | 0      |
| 859  | 860        | 3.668410 | 0.975118    | 0      |
| 4027 | 4028       | 3.522034 | 0.971308    | 1      |
| 3597 | 3598       | 3.266683 | 0.963268    | 1      |
| 3556 | 3557       | 3.243527 | 0.962440    | 1      |
| 5702 | 5703       | 3.241265 | 0.962358    | 1      |
| 2147 | 2148       | 3.241265 | 0.962358    | 0      |
| 36   | 37         | 3.241265 | 0.962358    | 0      |
| 3094 | 3095       | 3.241265 | 0.962358    | 0      |
| 4742 | 4743       | 3.241265 | 0.962358    | 0      |
| 5475 | 5476       | 3.232248 | 0.962030    | 0      |
| 3461 | 3462       | 3.232248 | 0.962030    | 1      |

Total customers with caravan policies: 348

Baseline probability: 5.98%

Precision@100: 25.00%, Recall@100: 7.18%

Precision@200: 28.50%, Recall@200: 16.38%

Precision@500: 22.80%, Recall@500: 32.76%

Precision@1000: 16.40%, Recall@1000: 47.13%

## 1.4 Evaluation using test data

Evaluates how well the model performs on unseen customers. The model works nearly as well on new customers as on those used for training

```
[4]: # Load test data
test_df = pd.read_csv('tic_test.csv')
print(f"Loaded test data with {len(test_df)} customers")

# Calculate scores for test customers using the same model
def evaluate_on_test_data(test_df, top_drivers, top_pairs):
    print("Evaluating model on test data...")

    # Get prior probability from training data
    prior_probability = df['CARAVAN'].mean() # Using training data prior
    prior_odds = prior_probability / (1 - prior_probability)
    log_prior = math.log(prior_odds) if prior_odds > 0 else 0
```

```

# Initialize array for scores
test_scores = np.ones(len(test_df)) * log_prior

# Track features used to avoid duplicate counting
used_features = set()

# First use the top multiperspective pairs
for _, pair in top_pairs.iterrows():
    feature1 = pair['Feature1']
    value1 = pair['Value1']
    is_threshold1 = pair['Is_Threshold1']

    feature2 = pair['Feature2']
    value2 = pair['Value2']
    is_threshold2 = pair['Is_Threshold2']

    # Skip if either feature already used
    if feature1 in used_features or feature2 in used_features:
        continue

    # Add both features to the used set
    used_features.add(feature1)
    used_features.add(feature2)

    # Calculate likelihood ratios for this pair
    if is_threshold1:
        condition1 = test_df[feature1] >= value1
    else:
        condition1 = test_df[feature1] == value1

    if is_threshold2:
        condition2 = test_df[feature2] >= value2
    else:
        condition2 = test_df[feature2] == value2

    condition = condition1 & condition2

    # Use statistics from training data
    positive_instances = df['CARAVAN'].sum()
    negative_instances = len(df) - positive_instances

    # Count from training data
    train_condition1 = df[feature1] >= value1 if is_threshold1 else
    ↪ df[feature1] == value1
    train_condition2 = df[feature2] >= value2 if is_threshold2 else
    ↪ df[feature2] == value2
    train_condition = train_condition1 & train_condition2

```

```

# Calculate conditional probabilities with Laplace smoothing
count_with_C = ((df['CARAVAN'] == 1) & train_condition).sum() + 1
count_with_not_C = (train_condition & (df['CARAVAN'] == 0)).sum() + 1

P_X_given_C = count_with_C / (positive_instances + 2)
P_X_given_not_C = count_with_not_C / (negative_instances + 2)

# Calculate log likelihood ratio for this pair
log_ratio = math.log(P_X_given_C / P_X_given_not_C)

# Apply to matching test customers
test_scores[condition] += log_ratio

# For test customers that don't match this pair, apply the complement
↳ likelihood
comp_count_with_C = positive_instances - ((df['CARAVAN'] == 1) &
↳ train_condition).sum() + 1
comp_count_with_not_C = negative_instances - (train_condition &
↳ (df['CARAVAN'] == 0)).sum() + 1
comp_P_X_given_C = comp_count_with_C / (positive_instances + 2)
comp_P_X_given_not_C = comp_count_with_not_C / (negative_instances + 2)
comp_log_ratio = math.log(comp_P_X_given_C / comp_P_X_given_not_C)

test_scores[~condition] += comp_log_ratio

# Then add top uniperspective drivers for features not yet used
for _, row in top_drivers.iterrows():
    desc = row['Descripción']
    is_threshold = '>=' in desc

    if is_threshold:
        parts = desc.split(' >= ')
        feature = parts[0]
        value = int(parts[1])
    else:
        parts = desc.split(' = ')
        feature = parts[0]
        value = int(parts[1])

    # Skip if feature already used in pairs
    if feature in used_features:
        continue

    # Add feature to used set
    used_features.add(feature)

```

```

# Create condition for test data
if is_threshold:
    condition = test_df[feature] >= value
else:
    condition = test_df[feature] == value

# Create condition for training data (to get statistics)
if is_threshold:
    train_condition = df[feature] >= value
else:
    train_condition = df[feature] == value

# Count from training data
positive_instances = df['CARAVAN'].sum()
negative_instances = len(df) - positive_instances

# Calculate conditional probabilities with Laplace smoothing
count_with_C = ((df['CARAVAN'] == 1) & train_condition).sum() + 1
count_with_not_C = (train_condition & (df['CARAVAN'] == 0)).sum() + 1

P_X_given_C = count_with_C / (positive_instances + 2)
P_X_given_not_C = count_with_not_C / (negative_instances + 2)

# Calculate log likelihood ratio
log_ratio = math.log(P_X_given_C / P_X_given_not_C)

# Apply to matching test customers
test_scores[condition] += log_ratio

# For test customers that don't match, apply the complement likelihood
comp_count_with_C = positive_instances - ((df['CARAVAN'] == 1) &
↳train_condition).sum() + 1
comp_count_with_not_C = negative_instances - (train_condition &
↳(df['CARAVAN'] == 0)).sum() + 1
comp_P_X_given_C = comp_count_with_C / (positive_instances + 2)
comp_P_X_given_not_C = comp_count_with_not_C / (negative_instances + 2)
comp_log_ratio = math.log(comp_P_X_given_C / comp_P_X_given_not_C)

test_scores[~condition] += comp_log_ratio

# Create DataFrame with customer IDs and scores
scores_df = pd.DataFrame({
    'CustomerID': test_df['NID'],
    'Score': test_scores,
    'Probability': 1 / (1 + np.exp(-test_scores)) # Convert log-odds to
↳probability
})

```

```

# Add actual values if they exist in the test data
if 'CARAVAN' in test_df.columns:
    scores_df['Actual'] = test_df['CARAVAN']

# Sort by score in descending order
scores_df = scores_df.sort_values(by='Score', ascending=False)

return scores_df

# Evaluate on test data
test_scores = evaluate_on_test_data(test_df, top_uniperspective_for_scoring,
    ↪top_multiperspective_for_scoring)

# Display the top 20 test customers most likely to purchase
print("\nTop 20 Test Customers Most Likely to Purchase Mobile Home Insurance:")
top_test_customers = test_scores.head(20)

# Show prediction results
if 'Actual' in test_scores.columns:
    print(top_test_customers[['CustomerID', 'Score', 'Probability', 'Actual']])

# Calculate test performance metrics
total_positive_test = test_df['CARAVAN'].sum()
print(f"\nTest data statistics:")
print(f"Total test customers with caravan policies: {total_positive_test}")
print(f"Test baseline probability: {(total_positive_test / len(test_df)) * 100:.2f}%")
    ↪100:.2f}%)

# Calculate precision at different thresholds
for k in [100, 200, 500, 1000]:
    if len(test_scores) >= k:
        precision = test_scores.head(k)['Actual'].sum() / k
        recall = test_scores.head(k)['Actual'].sum() / total_positive_test
        print(f"Test Precision@{k}: {precision*100:.2f}%, Test Recall@{k}: 100:.2f}%")
    ↪{recall*100:.2f}%)

# Calculate lift for the top N groups
for n in [100, 200, 300, 500, 1000]:
    if len(test_scores) >= n:
        top_n = test_scores.head(n)
        lift = (top_n['Actual'].mean() / (total_positive_test / len(test_df)))
    ↪len(test_df)))
        print(f"Test Lift for top {n} customers: {lift:.2f}x")

# Compare training vs test performance
print("\nComparison of Training vs Test Performance:")

```

```

train_precision_100 = customer_scores.head(100)['Actual'].sum() / 100
test_precision_100 = test_scores.head(100)['Actual'].sum() / 100
print(f"Precision@100: Training={train_precision_100*100:.2f}%,  

↳Test={test_precision_100*100:.2f}%")
else:
    print(top_test_customers[['CustomerID', 'Score', 'Probability']])
    print("\nNote: Test data does not contain target variable (CARAVAN), so  

↳performance metrics cannot be calculated.")

```

Loaded test data with 4000 customers

Evaluating model on test data...

Top 20 Test Customers Most Likely to Purchase Mobile Home Insurance:

|      | CustomerID | Score    | Probability | Actual |
|------|------------|----------|-------------|--------|
| 947  | 948        | 5.814319 | 0.997024    | 1      |
| 2587 | 2588       | 4.663669 | 0.990656    | 0      |
| 3482 | 3483       | 4.382899 | 0.987665    | 0      |
| 2388 | 2389       | 4.382899 | 0.987665    | 0      |
| 2047 | 2048       | 4.079688 | 0.983369    | 0      |
| 3772 | 3773       | 3.668410 | 0.975118    | 0      |
| 2621 | 2622       | 3.439944 | 0.968930    | 1      |
| 3726 | 3727       | 3.241265 | 0.962358    | 1      |
| 364  | 365        | 3.241265 | 0.962358    | 0      |
| 2416 | 2417       | 3.232248 | 0.962030    | 1      |
| 2164 | 2165       | 3.232248 | 0.962030    | 1      |
| 2634 | 2635       | 3.232248 | 0.962030    | 1      |
| 1242 | 1243       | 3.232248 | 0.962030    | 0      |
| 2843 | 2844       | 3.232248 | 0.962030    | 0      |
| 1467 | 1468       | 3.232248 | 0.962030    | 1      |
| 2118 | 2119       | 3.232248 | 0.962030    | 1      |
| 3842 | 3843       | 3.232248 | 0.962030    | 0      |
| 2132 | 2133       | 3.232248 | 0.962030    | 0      |
| 2515 | 2516       | 3.232248 | 0.962030    | 1      |
| 3347 | 3348       | 3.232248 | 0.962030    | 0      |

Test data statistics:

Total test customers with caravan policies: 238

Test baseline probability: 5.95%

Test Precision@100: 21.00%, Test Recall@100: 8.82%

Test Precision@200: 17.00%, Test Recall@200: 14.29%

Test Precision@500: 15.60%, Test Recall@500: 32.77%

Test Precision@1000: 12.20%, Test Recall@1000: 51.26%

Test Lift for top 100 customers: 3.53x

Test Lift for top 200 customers: 2.86x

Test Lift for top 300 customers: 3.03x

Test Lift for top 500 customers: 2.62x

Test Lift for top 1000 customers: 2.05x

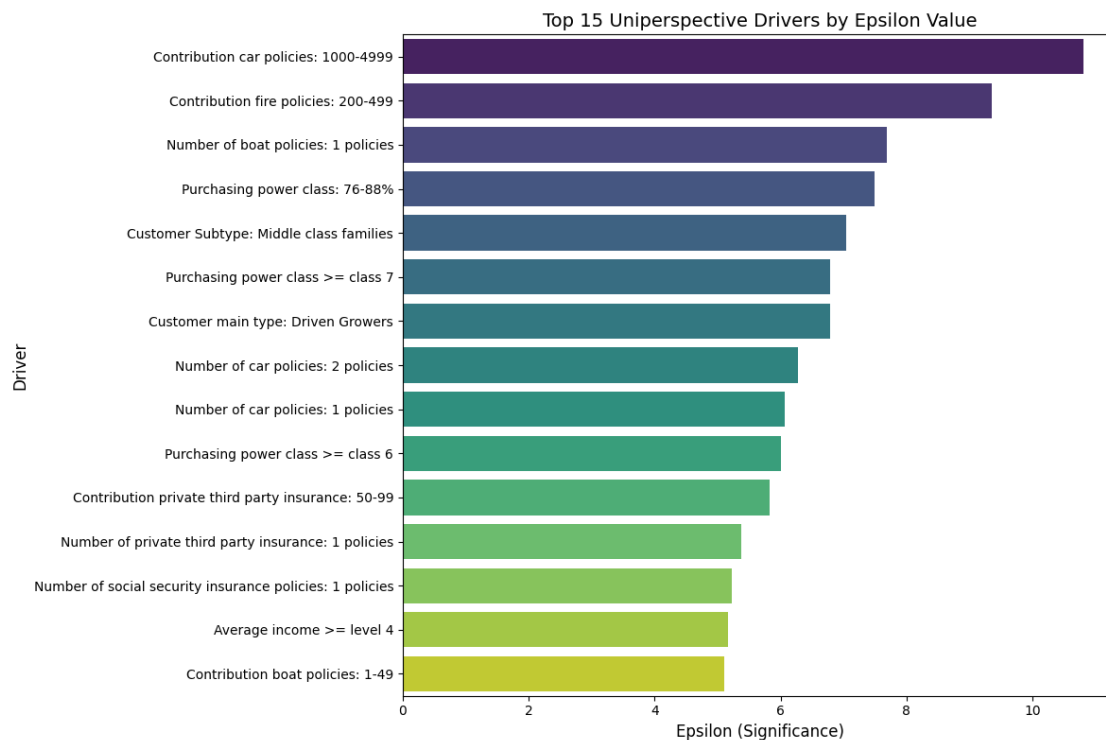
Comparison of Training vs Test Performance:  
Precision@100: Training=25.00%, Test=21.00%

## 1.5 Plots

Visualization of epsilon scores of uniperspective drivers

```
[5]: import matplotlib.pyplot as plt
import seaborn as sns

# Plot top uniperspective drivers
plt.figure(figsize=(12, 8))
sns.barplot(x='Epsilon', y='Driver', data=output_df.head(15),
            palette='viridis', hue='Driver', legend=False)
plt.title('Top 15 Uniperspective Drivers by Epsilon Value', fontsize=14)
plt.xlabel('Epsilon (Significance)', fontsize=12)
plt.ylabel('Driver', fontsize=12)
plt.tight_layout()
plt.show()
```



Lift charts (lift = how much better is the model compared to a random selection), depending on the number of customers selected.

```

[10]: # Create data for lift chart - both training and test
lifts_train = []
lifts_test = []
percentiles = [100, 200, 300, 500, 1000, 2000, 3000, 5000]

# Calculate baseline rates
baseline_train = df['CARAVAN'].mean()
baseline_test = test_df['CARAVAN'].mean()

# Calculate lifts for training data
for n in percentiles:
    if n <= len(customer_scores):
        lift = customer_scores.head(n)['Actual'].mean() / baseline_train
        lifts_train.append(lift)

# Calculate lifts for test data
for n in percentiles:
    if n <= len(test_scores):
        lift = test_scores.head(n)['Actual'].mean() / baseline_test
        lifts_test.append(lift)

# Plot lift chart with both datasets
plt.figure(figsize=(12, 7))

# Training data line
plt.plot(percentiles[:len(lifts_train)], lifts_train, 'o-',
         linewidth=2, markersize=8, color='#2C7BB6', label='Training Data')

# Test data line
plt.plot(percentiles[:len(lifts_test)], lifts_test, 's-',
         linewidth=2, markersize=8, color='#D7191C', label='Test Data')

plt.axhline(y=1, color='gray', linestyle='--', alpha=0.7, label='Baseline_
↳ (Random Selection)')

# Enhance the chart
plt.title('Lift Chart: Model Performance Comparison', fontsize=15)
plt.xlabel('Number of Customers Targeted', fontsize=13)
plt.ylabel('Lift', fontsize=13)
plt.grid(True, alpha=0.3)
plt.legend(fontsize=12)

# Add value annotations at key points
for i, (x, y) in enumerate(zip(percentiles[:len(lifts_train)], lifts_train)):
    if i % 2 == 0: # Annotate every other point to avoid clutter
        plt.annotate(f'{y:.2f}x', (x, y), textcoords="offset points",
                     xytext=(0,10), ha='center', fontsize=9, color='#2C7BB6')

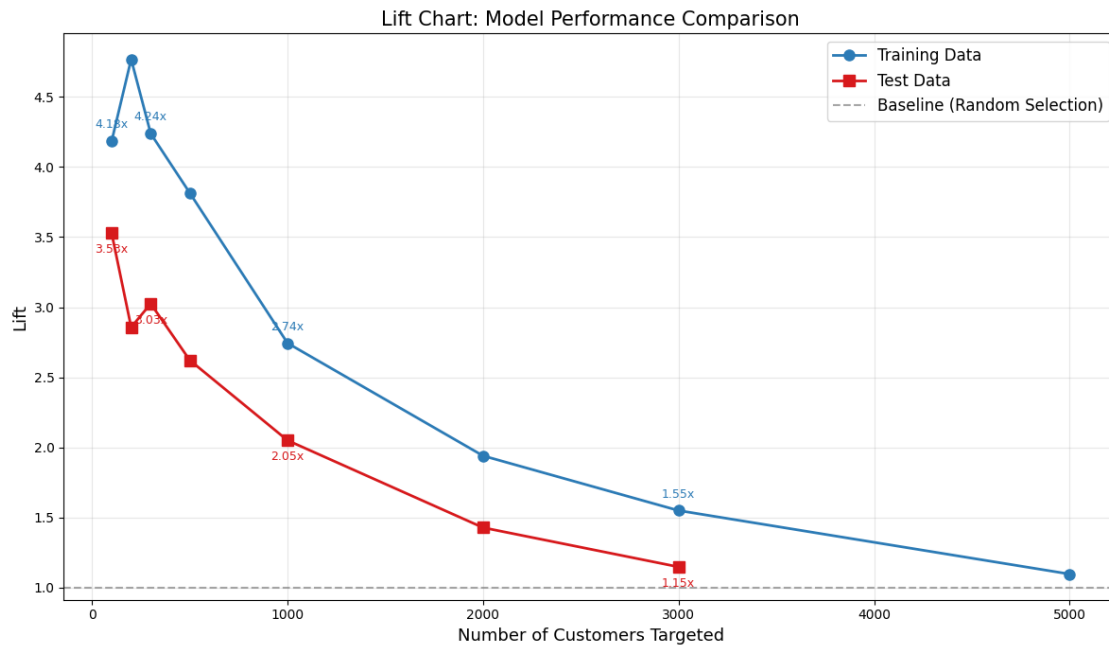
```

```

for i, (x, y) in enumerate(zip(percentiles[:len(lifts_test)], lifts_test)):
    if i % 2 == 0: # Annotate every other point
        plt.annotate(f'{y:.2f}x', (x, y), textcoords="offset points",
                     xytext=(0,-15), ha='center', fontsize=9, color='#D7191C')

plt.tight_layout()
plt.show()

```



Comparison of how well the model performs (Precision@k) for training and test data.

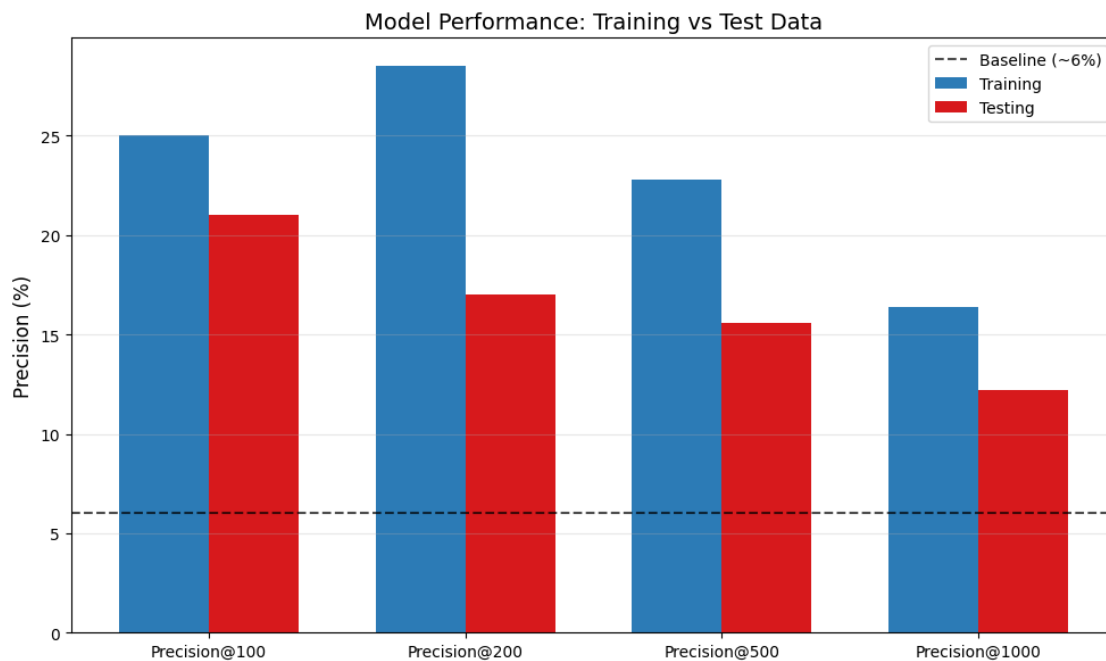
```

[7]: # Comparison data
metrics = ['Precision@100', 'Precision@200', 'Precision@500', 'Precision@1000']
training = [25.0, 28.5, 22.8, 16.4] # From your output
testing = [21.0, 17.0, 15.6, 12.2] # From your output

# Create bar chart
plt.figure(figsize=(10, 6))
x = range(len(metrics))
width = 0.35
plt.bar([i - width/2 for i in x], training, width, label='Training',
        color='#2C7BB6')
plt.bar([i + width/2 for i in x], testing, width, label='Testing',
        color='#D7191C')
plt.axhline(y=6.0, color='black', linestyle='--', alpha=0.7, label='Baseline
        (~6%)')

```

```
plt.xticks(x, metrics)
plt.title('Model Performance: Training vs Test Data', fontsize=14)
plt.ylabel('Precision (%)', fontsize=12)
plt.legend()
plt.grid(True, alpha=0.3, axis='y')
plt.tight_layout()
plt.show()
```



The cumulative gains chart displays the percentage of total mobile home insurance purchasers that can be identified when targeting different percentages of customers based on model scores, demonstrating how much more efficiently the model performs compared to random selection.

```
[9]: # Calculate cumulative response
def calculate_cumulative_response(scores_df, total_population):
    cumulative = []
    purchasers_found = 0
    step_size = 100

    for i in range(step_size, len(scores_df) + 1, step_size):
        purchasers_found += scores_df.iloc[i-step_size:i]['Actual'].sum()
        cumulative.append(purchasers_found / total_population * 100)

    return cumulative

population_pcts = [i for i in range(0, 101, 5)]
random_selection = population_pcts.copy() # Diagonal line
```

```

# Calculate actual cumulative responses (percentage of total purchasers found)
steps = [int(len(customer_scores) * p/100) for p in population_pcts]
cumulative_pct = [customer_scores.head(n)['Actual'].sum() / total_positive * 100
                  if n > 0 else 0 for n in steps]

# Plot
plt.figure(figsize=(10, 8))
plt.plot(population_pcts, cumulative_pct, 'b-', linewidth=2, label='Model')
plt.plot(population_pcts, random_selection, 'r--', linewidth=2, label='Random_
↳Selection')
plt.title('Cumulative Gains Chart', fontsize=14)
plt.xlabel('Percentage of Customers Targeted', fontsize=12)
plt.ylabel('Percentage of Total Purchasers Found', fontsize=12)
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()

```

